

Lab 3: Component Modelling & Architectural Pattern Selection

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Problem Statement #09: Faculty Research Grant & Publication Tracker

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker System.

Architectural Choice

The system uses a three-layer architecture: a Presentation Layer for the faculty and dean portal, a Business Layer for grant and publication logic, and a Data Layer for persistent storage and audit records.

Reason 1

Separation of concerns lets the Research Dean's burn-up dashboard (FR-004) and milestone alerting (FR-005) evolve independently from the core grant approval and expense rules in the Business Layer. Changes to the UI never risk breaking financial logic.

Reason 2

Layering isolates the dependency on the External Citation DB. When the DOI lookup is slow or unreachable, as described in Alternate Flow 5a of the use case spec, the Business Layer retries and falls back to a manual flag without the Presentation Layer or Data Layer being affected.

Security Advantage

The Audit Ledger Component sits in the Data Layer and is only reachable through the Grant and Publication Manager in the Business Layer. No component outside that path can write to it directly, which supports the tamper-evident, immutable audit requirement in NFR-001.

Performance Benefit

Citation counts fetched through the Citation Fetch Service can be cached in the Business Layer instead of re-fetched on every dashboard refresh. This keeps the burn-up dashboard update lag under the 5-minute target in FR-004 and reduces load on the external service.